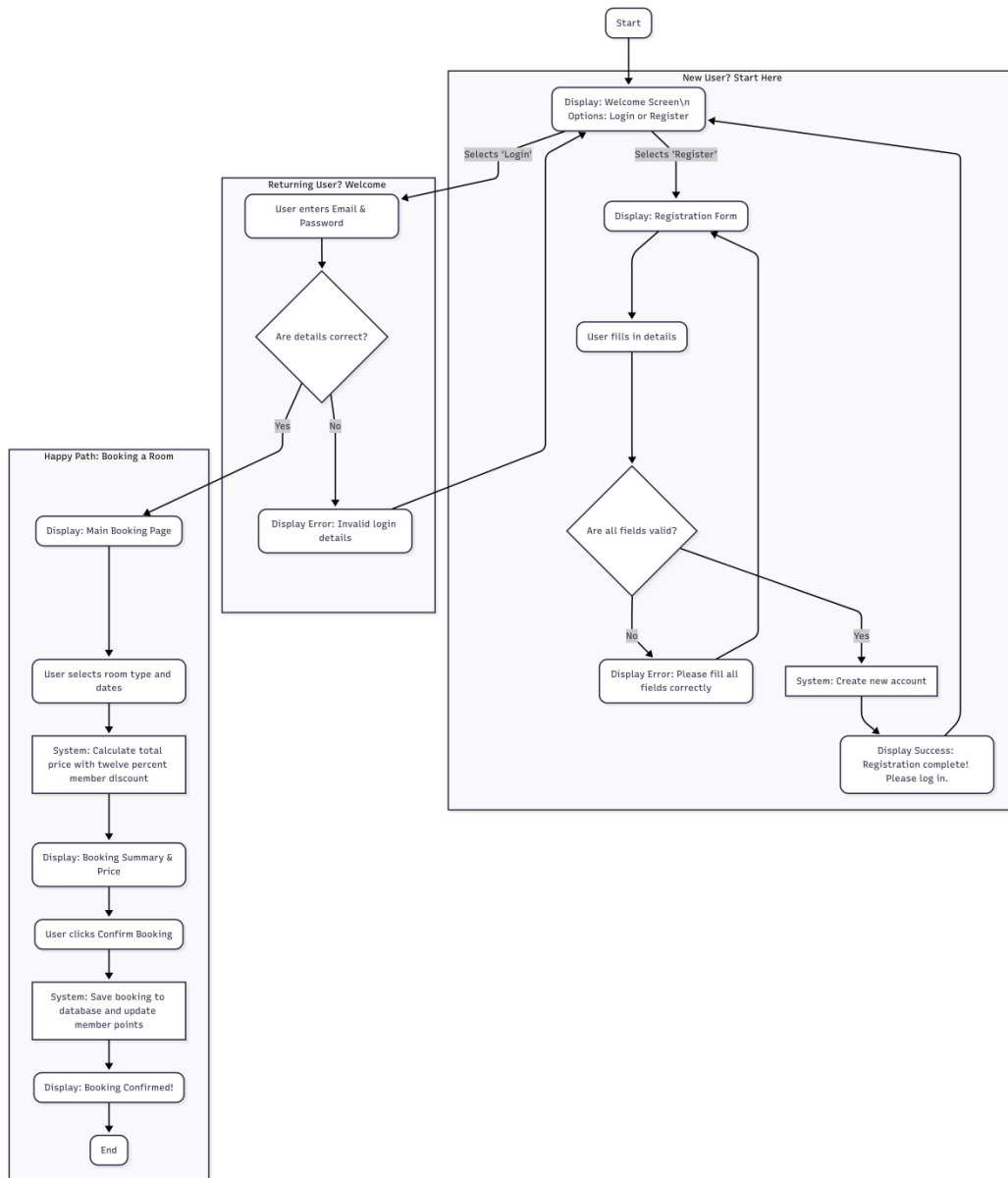




**KUALA LUMPUR UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

(formerly known as Infrastructure University Kuala Lumpur)

Project Title:	Hotel Booking Management System
Course Code:	CSC1513 / CSC1514
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Submission Date:	07/10/2025

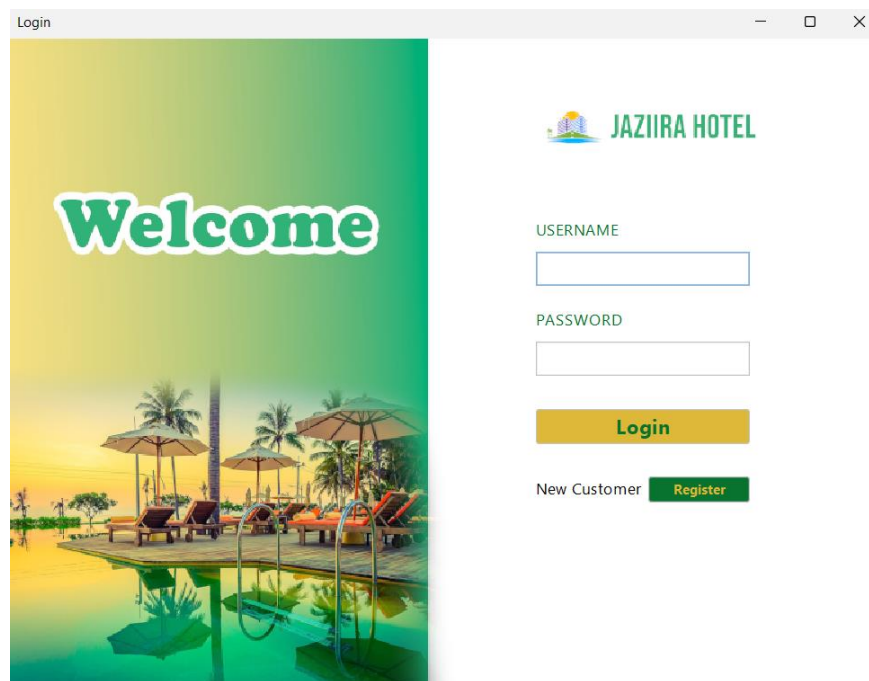


Our project is a **Hotel Booking Management System**. The main goal is to create a simple and easy-to-use program with a graphical user interface (GUI) for booking hotel rooms. The system is designed for general users and registered members.

The key features of our system are:

- **Customer Registration and Login:** New users can create an account by providing their personal details. Existing users can log in with their email and password.
- **Room Booking:** After logging in, users can see the available room types: Single, Double, and Suite, with their prices per night. They can select a room and choose their check-in and check-out dates.
- **Membership Benefits:** All registered members receive a **12% discount** on their bookings. They also earn **1 loyalty point for every RM10 spent**.
- **Booking Summary:** The system automatically calculates the total price, including the member discount, and shows a full summary on a confirmation page before the booking is final.
- **Technology:** We used the **Java programming language** and the **Java Swing library** to build the graphical user interface. We used an in-memory list to store customer data, as required.

1. Feature: User Login



```
118 private void btnLoginActionPerformed(java.awt.event.ActionEvent evt) {  
119     // TODO add your handling code here:  
120     String email = tfEmail.getText();  
121     String password = new String(pfPassword.getPassword());  
122     User loggedInUser = null;  
123  
124     // Search for the user in your stored user list  
125     for (User user : UserData.userList) {  
126         if (user.getEmail().equals(email) && user.getPassword().equals(password)) {  
127             loggedInUser = user; // Found the user!  
128             break;  
129         }  
130     }  
131  
132     // Check if the user was found  
133     if (loggedInUser != null) {  
134         // SUCCESS: Open the main booking page  
135         JOptionPane.showMessageDialog(this, "Welcome, " + loggedInUser.getName() + "!");  
136         new room(loggedInUser).setVisible(true); // Opens the room booking window  
137         this.dispose(); // Closes the login window  
138     } else {  
139         // FAILURE: Show an error message  
140         JOptionPane.showMessageDialog(this, "Invalid email or password.", "Login Failed", JOptionPane.ERROR_MESSAGE);  
141     }  
142 }
```

Code Explanation:

This code runs when the "Login" button is clicked.

- It gets the email and password that the user typed into the text fields.
- It then loops through the list of registered users stored in memory.
- Inside the loop, it checks if the entered email and password match a user in the list.
- If a match is found, it opens the main room booking page and closes the login window.
- If no match is found after checking all users, it shows an "Invalid email or password" error message.

2. Feature: User Registration

Register

Full Name

Contact No

Email address

ID/Passport

Password

Register

Have already registered Login

JAZIIRA HOTEL

```
143 private void jButtonActionPerformed(java.awt.event.ActionEvent evt) {  
144     // TODO add your handling code here:  
145  
146     // Get the text from all the form fields  
147     String name = tfName.getText();  
148     String email = tfEmail.getText();  
149     String password = new String(pfPassword.getPassword());  
150     String contact = tfContact.getText();  
151     String id = tfId.getText();  
152  
153     // 1. Basic Validation to make sure fields are not empty  
154     if (name.isEmpty() || email.isEmpty() || password.isEmpty()) {  
155         JOptionPane.showMessageDialog(this, "Please fill in all required fields.", "Error", JOptionPane.ERROR_MESSAGE);  
156         return; // Stop the process if validation fails  
157     }  
158  
159     // 2. Check if a user with this email already exists  
160     for (User existingUser : UserData.userList) {  
161         if (existingUser.getEmail().equalsIgnoreCase(email)) {  
162             JOptionPane.showMessageDialog(this, "A user with this email already exists.", "Error", JOptionPane.ERROR_MESSAGE);  
163             return; // Stop if the email is a duplicate  
164         }  
165     }  
166  
167     // 3. If everything is okay, create the new user  
168     User newUser = new User(name, email, password, contact, id);  
169     UserData.userList.add(newUser);  
170  
171     // 4. Show a success message and return to the login screen  
172     JOptionPane.showMessageDialog(this, "Registration Successful! Please log in.");  
173     new login().setVisible(true);  
174     this.dispose();  
175 }
```

Code Explanation: This code runs when the "Register" button is clicked on the registration page.

- It gets all the details the user entered in the form (name, email, password, etc.).
- It checks to make sure the most important fields are not empty.
- It then creates a new User object with these details.
- The new

User object is added to the program's memory list, saving the new account.

- Finally, it shows a "Registration Successful" message and takes the user back to the login page.

3. Feature: Room Booking and Price Calculation

Room Booking





Room Type : Single
Price : RM150/night
☐ Select



Room Type : double
Price : RM250/night
☐ Select



Room Type : Suite
price : RM450/night
☐ Select

Check-in Date

Check-out Date

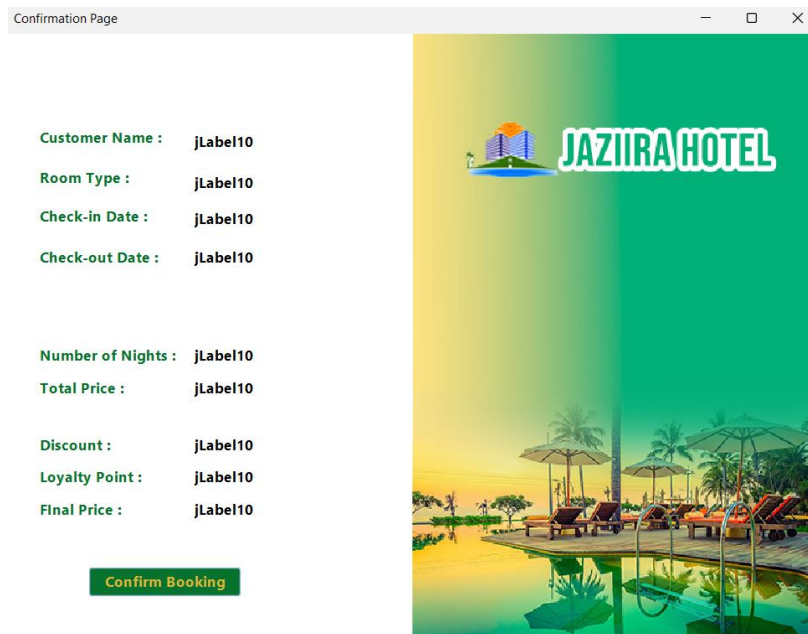
Book Now

```
184 private void jButtonActionPerformed(java.awt.event.ActionEvent evt) {
185     // TODO add your handling code here:
186
187     try {
188         // 1. Get the selected room and its price from your JRadioButtons
189         String selectedRoomType = "";
190         double pricePerNight = 0;
191
192         if (jRadioButton1.isSelected()) {
193             selectedRoomType = "Single";
194             pricePerNight = 150;
195         } else if (jRadioButton2.isSelected()) {
196             selectedRoomType = "Double";
197             pricePerNight = 250;
198         } else if (jRadioButton3.isSelected()) {
199             selectedRoomType = "Suite";
200             pricePerNight = 400;
201         } else {
202             JOptionPane.showMessageDialog(this, "Please select a room type.", "Error", JOptionPane.ERROR_MESSAGE);
203             return; // Stop if no room is selected
204         }
205
206         // 2. Calculate the number of nights from your JDateChoosers
207         if (jDateChooser1.getDate() == null || jDateChooser2.getDate() == null) {
208             JOptionPane.showMessageDialog(this, "Please select both a check-in and check-out date.", "Error", JOptionPane.ERROR_MESSAGE);
209             return; // Stop if dates are not selected
210         }
211
212         // Convert dates for calculation
213         java.time.LocalDate checkInDate = jDateChooser1.getDate().toInstant().atZone(java.time.ZoneId.systemDefault()).toLocalDate();
214         java.time.LocalDate checkOutDate = jDateChooser2.getDate().toInstant().atZone(java.time.ZoneId.systemDefault()).toLocalDate();
215         long nights = java.time.temporal.ChronoUnit.DAYS.between(checkInDate, checkOutDate);
216
217         if (nights <= 0) {
218             JOptionPane.showMessageDialog(this, "Check-out date must be after check-in date.", "Date Error", JOptionPane.ERROR_MESSAGE);
219             return;
220         }
221     }
```

Code Explanation: This code runs when the "Book Now" button is clicked.

- It checks which room type the user selected to get the price per night (e.g., RM150 for Single, RM250 for Double).
- It gets the check-in and check-out dates and calculates the total number of nights.
- It then calculates the baseCost, the member discount (12%) , the totalCost, and the pointsEarned (1 point for every RM10 spent).
- Finally, it opens the confirmationpage and sends all these calculated details to it for display.

4.Feature: Displaying the Booking Confirmation



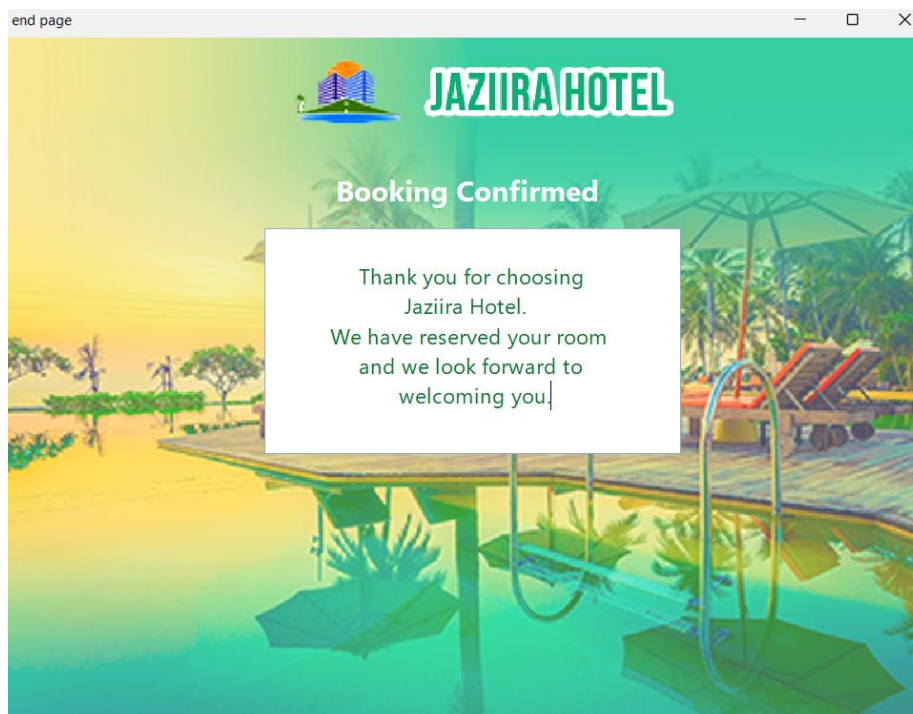
```
22 public confirmationpage(User user, String roomType, String checkIn, String checkOut, long nights, double totalPrice, double discount, int points, double finalPrice) {
23     initComponents(); // This loads your GUI design
24
25     // This code sets the text of each JLabel
26     lblName.setText(user.getName());
27     lblRoomType.setText(roomType);
28     lblCheckIn.setText(checkIn);
29     lblCheckOut.setText(checkOut);
30     lblNights.setText(String.valueOf(nights));
31     lblTotalPrice.setText(String.format("RM %.2f", totalPrice));
32     lblDiscount.setText(String.format("-RM %.2f", discount));
33     lblLoyalty.setText(String.valueOf(points));
34     lblFinalPrice.setText(String.format("RM %.2f", finalPrice));
35 }
36
```

Code Explanation: This code is a special constructor that runs when the confirmationpage is created.

- It receives all the booking details (the user, room type, dates, costs, etc.) from the previous room page.
- It then uses the

.setText() method to display each piece of information in the correct JLabel on the screen. This creates the final booking summary for the user to see.

5. Feature: Final Confirmation End Page



```
188 |
189 | private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
190 |
191 |     // TODO add your handling code here:
192 |     {setVisible(false);
193 |     new end().setVisible(true); }
194 |
195 | }
196 |
```

Code Explanation: This code runs when the "Confirm Booking" button is clicked on the confirmationpage.

- The line `new end().setVisible(true);` creates and shows the final "Thank You" window.
- The line `this.dispose();` closes the confirmation page.
- This is the final step in the booking process, confirming to the user that their booking is complete.